

A non-linear model predictive controller with obstacle avoidance for a space robot

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Abstract

This study investigates the use of the non-linear model predictive control (NMPC) strategy for a kinematically redundant space robot to approach an un-cooperative target in complex space environment. Collision avoidance, traditionally treated as a high level planning problem, can be effectively translated into control constraints as part of the NMPC. The objective of this paper is to evaluate the performance of the predictive controller in a constrained workspace and to investigate the feasibility of imposing additional constraints into the NMPC. In this paper, we reformulated the issue of the space robot motion control by using NMPC with predefined objectives under input, output and obstacle constraints over a receding horizon. An on-line quadratic programming (QP) procedure is employed to obtain the constrained optimal control decisions in real-time. This study has been implemented for a 7 degree-of-freedom (DOF) kinematically redundant manipulator mounted on a 6 DOF free-floating spacecraft via simulation studies. Real-time trajectory tracking and collision avoidance particularly demonstrate the effectiveness and potential of the proposed NMPC strategy for the space robot.

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1. Introduction

The existence of the space debris limits the future space activities and increases the risk of the operative satellites. How to remove these impeditive factors received fairly significant attentions. The application of space robots to perform such task is attractive as a result of their versatility, flexibility and expandability. Over the past few decades, series of researches have been conducted in the field of space robots. Examples include “Robot Technology Experiment (ROTEX)” (Hirzinger et al., 1994),

“Engineering Test Satellite VII (ETS-VII)” (Oda et al., 1996) and “Orbital Express (OE)” (Ogilvie et al., 2008). In light of the space robots currently planned by world wide space agencies, an increase in the number and capacity of the robots applied in space missions will be a foregone conclusion in the coming future (Rekleitis et al., 2007).

In order to perform manipulation near space debris, how to control the motion of a robotic manipulator in its workspace with satisfactory performance has been a prime concern over the past few decades. One of the most widely used method for the space robots is the resolved motion acceleration control (RMAC) as developed in Umetani and Yoshida (1989) and Papadopoulos and Moosavian (1995). It employs a mathematical model of the space robot for dynamics compensation. Another widespread control technique is adaptive control as introduced in Xu et al.

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(1992) and Abiko and Hirzinger (2009). However, above control strategies own such deficiencies: inability to solve system input and output boundaries, lack of optimization, and incapability to deal with additional constraints. Originated from the chemical processing industries, model predictive control (MPC) (Qin and Badgwell, 2003) has gradually expanded its application to the other field, such as in Lin and Liu (2012), McCourt and de Silva (2006) and Jasour and Farrokhi (2009). However, the practical application of MPC in the field of space robotics is still rare. Another inevitable issue during space manipulator's motion is collision avoidance. Conventionally, it is treated as a high level planning problem which is independent from the design of the control strategy. Numerous solutions, like global planning methods (Galicki, 1992; Stilman, 2010), harmonic potential functions (Kim and Khosla, 1992), Jacobian null-space (Glass et al., 1995), position control (Seraji and Bon, 1999), Jacobian transpose method (Lee and Buss, 2007) and constrained local optimization (Zhang and Wang, 2004; Kanoun et al., 2011) have been developed to solve such issue during path planning. Considering the application background of space debris removal, the shortages of the traditional control methods and the demands of real-time collision avoidance motivate us to seek new methods for space robots to meet the specific space missions.

In this paper, we propose a general control framework with non-linear model predictive control (NMPC) applied to a kinematically redundant space manipulator considering collision avoidance and joint physical limits. The reason for choosing a kinematically redundant manipulator is the redundancy resolution can be employed for additional objectives, such as minimize base disturbance, avoid collision, or maximize the manipulability, and so forth. The fundamental idea of this paper is to separate the realization of the predefined task, depicted by the minimization of cost function, from the constraints of input, output and anti-collision. Inspired by the velocity damper method (Kanehiro et al., 2009), anti-collision constraints can be initially translated into uniform inequality conditions, then integrated into a quadratic programming (QP) procedure to obtain an optimal resolution over the receding horizon. The method proposed in this paper shows it is a bridge connecting the upper path planning level and lower actuator's control level, moreover, it handles the multi-variable optimal control problems with constraints in a comprehensive and systematic way.

The rest of the paper is organized as follows. Section 2 presents the general dynamics model of a space robot. Through feedback linearization, a linearised model is derived from the highly non-linear space robot model. Section 3 illustrates the collision detection and the conversion of the anti-collision issue as the linear inequality constraints. Section 4 gives the NMPC design in detail regarding to the linearised model mentioned in Section 2. An overview of the NMPC, optimization index, inequality constraints handling and QP procedure are included in this

section. The simulation results are presented in Section 5. Our discussions, conclusions and future works are listed in the last section.

2. Modelling of a space robot

Before discussing the dynamics of a space robot in detail, some symbols and variables used in the following sections are listed in Table 1. A space robotic system is composed of a spacecraft and an n DOF manipulator, in total $n + 1$ bodies as shown in Fig. 1. Many investigations have been conducted in the field of the space robot dynamics. Refer to Wang and Walter (2013), the dynamics equations of a space robot using Lagrangian mechanism can be expressed as follows

$$\begin{bmatrix} \mathbf{H}_b & \mathbf{H}_{bm} \\ \mathbf{H}_{bm}^T & \mathbf{H}_m \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{x}}_b \\ \ddot{\boldsymbol{\theta}} \end{bmatrix} + \begin{bmatrix} \mathbf{c}_b \\ \mathbf{c}_m \end{bmatrix} = \begin{bmatrix} \mathbf{f}_b \\ \boldsymbol{\tau} \end{bmatrix} + \begin{bmatrix} \mathbf{J}_b^T \\ \mathbf{J}_e^T \end{bmatrix} \mathbf{f}_e \quad (1)$$

where $\ddot{\mathbf{x}}_b = (\ddot{\mathbf{r}}_b, \dot{\boldsymbol{\omega}}_b) \in \mathbb{R}^6$ is the vector of linear and angular accelerations of the base expressed in the inertial frame, $\ddot{\boldsymbol{\theta}} \in \mathbb{R}^n$ represents joint accelerations of the manipulator. If no external force and moment vectors applied to the end-effector, i.e. $\mathbf{f}_e = \mathbf{0}$ and no active actuators are applied to the base, i.e. $\mathbf{f}_b = \mathbf{0}$, then this is called a free-floating space robot. According to the angular momentum conservation law, the total momentum $\mathbf{L}_0 \in \mathbb{R}^6$ around the system center of mass is conserved in the free-floating mode, which can be expressed by

$$\mathbf{L}_0 = \mathbf{H}_b \dot{\mathbf{x}}_b + \mathbf{H}_{bm} \dot{\boldsymbol{\theta}} \quad (2)$$

Suppose the initial momentum $\mathbf{L}_0 = \mathbf{0}$, since $\mathbf{H}_b$ is always invertible, by substituting the motion of the base $\dot{\mathbf{x}}_b = -\mathbf{H}_b^{-1} \mathbf{H}_{bm} \dot{\boldsymbol{\theta}}$ into the kinematic mapping of the end-effector, $\dot{\mathbf{x}}_e = \mathbf{J}_b \dot{\mathbf{x}}_b + \mathbf{J}_e \dot{\boldsymbol{\theta}}$, the motion of the end-effector is given as follows

$$\dot{\mathbf{x}}_e = \begin{bmatrix} \dot{\mathbf{r}}_e \\ \boldsymbol{\omega}_e \end{bmatrix} = \mathbf{J}_e \dot{\boldsymbol{\theta}} = (\mathbf{J}_e - \mathbf{J}_b \mathbf{H}_b^{-1} \mathbf{H}_{bm}) \dot{\boldsymbol{\theta}} \quad (3)$$

Table 1
Kinematic and dynamic symbols used in the paper.

Symbols	Representation
J_i, C_i	Joint i and mass center of link i
$\mathbf{a}_i, \mathbf{b}_i \in \mathbb{R}^3$	Position vectors from J_i to C_i and from C_i to J_{i+1}
$\mathbf{r}_{C_i} \in \mathbb{R}^3$	Position vector of mass center of link i
$\mathbf{r}_b, \mathbf{r}_e \in \mathbb{R}^3$	Position vectors of base and end-effector
$\boldsymbol{\omega}_b, \boldsymbol{\omega}_e \in \mathbb{R}^3$	Angular velocities of base and end-effector
$\mathbf{I}_i \in \mathbb{R}^{3 \times 3}, m_i \in \mathbb{R}$	Inertia matrix and mass of link i
$\mathbf{H}_b \in \mathbb{R}^{6 \times 6}$	Inertia matrix of the base
$\mathbf{H}_{bm} \in \mathbb{R}^{6 \times n}$	Coupling inertia matrix between base and manipulator
$\mathbf{H}_m \in \mathbb{R}^{n \times n}$	Inertia matrix of the manipulator
$\mathbf{c}_b \in \mathbb{R}^6, \mathbf{c}_m \in \mathbb{R}^n$	Vectors of velocity dependent non-linear terms
$\mathbf{f}_b, \mathbf{f}_e \in \mathbb{R}^6$	Vectors of force and moment exert on base and end-effector
$\boldsymbol{\tau} \in \mathbb{R}^n$	Torque exert on manipulator joints

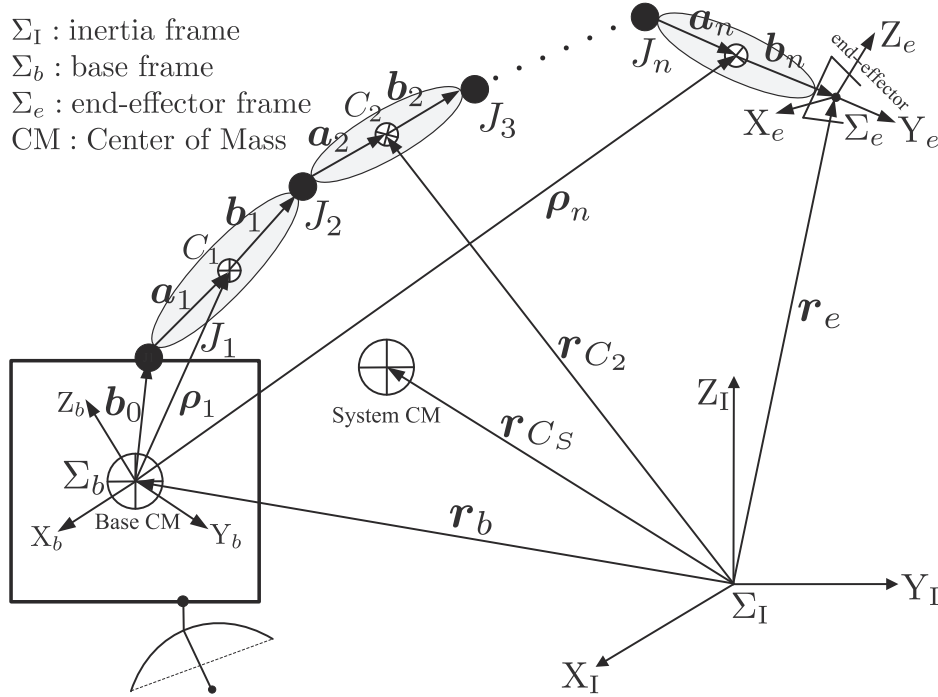


Fig. 1. Schematic diagram of a space robot.

$\mathbf{J}_g$ is termed Generalized Jacobian Matrix (GJM) which is derived in Umetani and Yoshida (1989). Consider Eq. (2), the dynamics formulation of a free-floating space robot can be described by

$$\tau = \mathbf{H}(\theta)\ddot{\theta} + \mathbf{c}(\theta, \dot{\theta}) \quad (4)$$

where $\mathbf{H}(\theta) = \mathbf{H}_m - \mathbf{H}_{bm}^T \mathbf{H}_b^{-1} \mathbf{H}_{bm} \in \mathbb{R}^{n \times n}$ and $\mathbf{c}(\theta, \dot{\theta}) = \mathbf{c}_m - \mathbf{H}_{bm}^T \mathbf{H}_b^{-1} \mathbf{c}_b \in \mathbb{R}^{n \times 1}$ are the generalized inertia matrix and the non-linear force and torque vectors of the free-floating space robot, respectively. Since the dynamic model in Eq. (4) is highly non-linear, therefore it is not possible to be a prediction model for commonly used MPC strategy. For the sake of deriving a linearised version of the space robot, a feedback linearization procedure (Siciliano et al., 2009, Ch. 8) is recalled here to obtain the state-space form. Taking $\mathbf{x} = [\theta, \dot{\theta}]^T$ as the system states and $\mathbf{u} = \ddot{\theta}$ as the new input, without considering the unmeasured noise, a discrete state-space model can be obtained utilizing the zero-order hold method with certain sampling interval h

$$\begin{cases} \mathbf{x}(k+1) = \mathbf{A}_d \mathbf{x}(k) + \mathbf{B}_d \mathbf{u}(k) \\ \mathbf{y}(k+1) = \mathbf{C}_d \mathbf{x}(k+1) \end{cases} \quad (5)$$

where $\mathbf{A}_d$, $\mathbf{B}_d$ and $\mathbf{C}_d$ are the system, input and output matrices in discrete-time, respectively. $\mathbf{x}(k)$, $\mathbf{u}(k)$ and $\mathbf{y}(k)$ are the states, input and measured output vectors at instant k , respectively. The NMPC controller requires the whole system states $\mathbf{x}(k)$ at each sampling time; however, all states in $\mathbf{x}(k)$ cannot be obtained directly from measurements. In this paper, a Kalman estimator (Wang, 2009, Ch. 1) is implemented to estimate the full system states.

3. Anti-collision constraints

The existence of the obstacles in the workspace of a space manipulator is undesired, since it restricts the motion of the manipulator. Hence, possible collisions must be concerned during the motion of the manipulator. Practically, the obstacles can be detected with the sensors, such as lidar or stereo camera. The detected obstacles can be enveloped by tight-fitting bounding volumes to minimize the computational requirements. Before we go into the collision avoidance algorithm in detail, let us first recall the definition of the strictly convex objects.

Strictly convex objects. Let $\mathcal{O}$ represent a closed subset of $\mathbb{R}^3$ and $X(\mathcal{O})$ be the interior of (greater open subset of) $\mathcal{O}$. $\mathcal{O}$ is strictly convex if and only if $\lambda A + (1 - \lambda)B \in X(\mathcal{O})$, $\forall A \in \mathcal{O}$, $\forall B \in \mathcal{O}$ and $0 < \lambda < 1$.

As shown in Fig. 2, let $\mathcal{O}_1$ and $\mathcal{O}_2$ stand for the tight-fitting bounding volumes of two movable objects and P_1 and $P_{1'}$ denote the closest points between $\mathcal{O}_1$ and $\mathcal{O}_2$. Since both objects are strictly convex, the distance $d = \|P_1 P_{1'}\|$ is continuously differentiable. The derivative of the distance $\|P_1 P_{1'}\|$ can be expressed by the following equation

$$\dot{d} = \langle \dot{\mathbf{r}}_{P_1} - \dot{\mathbf{r}}_{P_{1'}}, \mathbf{n}_1 \rangle \quad (6)$$

where $\mathbf{r}_{P_i}$ and $\dot{\mathbf{r}}_{P_i}$ represent vectors of inertial position and velocity of point P_i . $\mathbf{n}_1 = (\mathbf{r}_{P_1} - \mathbf{r}_{P_{1'}})/d$ is a unit vector and the notation $\langle \mathbf{u}, \mathbf{v} \rangle$ refers to the inner product of vector $\mathbf{u}$ and $\mathbf{v}$. Let $\mathbf{q} = [\mathbf{x}_b, \boldsymbol{\theta}]^T$ be the configuration and $\dot{\mathbf{q}}$ the generalized velocity of space robot, the task-space velocity of P_i can therefore be denoted by $\dot{\mathbf{r}}_{P_i} = \mathbf{J}_{P_i} \dot{\mathbf{q}}$, where $\mathbf{J}_{P_i} \in \mathbb{R}^{3 \times (n+6)}$ is a Jacobian matrix of $\mathcal{O}_1$ at point P_i .

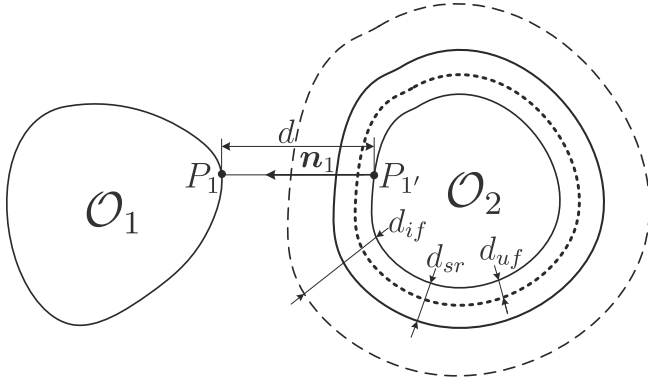


Fig. 2. The relationship between two objects.

Let us recall an inequality that is named velocity damper proposed in Kanehiro et al. (2009), if d enters into a threshold called influence distance expressed by d_{if} , the following constraint is defined

$$\dot{d} \geq -\xi \frac{d - d_{uf}}{d_{if} - d_{uf}} \quad (7)$$

where $\xi > 0$ is for adjusting convergence speed, $d_{uf} < d_{if}$ is a positive quantity named unsafe distance. Given the initial condition $d(0) \geq d_{uf}$, the following inequality can be derived from Eq. (4)

$$d(t) \geq d_{uf} + (d(0) - d_{uf})e^{\frac{-\xi}{d_{if} - d_{uf}}t} > d_{uf}, \quad \forall t > 0 \quad (8)$$

Eq. (8) assures that the minimum distance between the two objects constrained by the velocity damper will be never smaller than d_{uf} . Similarly, $d_{uf} < d_{sr} < d_{if}$ defines another quantity named security distance which indicates that once $d \leq d_{sr}$, the inequality constraint expressed in Eq. (7) will become a hard constraint. Note that $\dot{d}$ is constrained by the inequality (7), when P_1 enters into the influence zone ($d \leq d_{if}$), plug Eq. (6) into inequality (7), a linear inequality constraint over the generalized velocity $\dot{\mathbf{q}}$ can be obtained

$$\langle \mathbf{J}_{P_1}^T \mathbf{n}_1, \dot{\mathbf{q}} \rangle \geq \langle \dot{\mathbf{r}}_{P_1'}, \mathbf{n}_1 \rangle - \xi \frac{d - d_{uf}}{d_{if} - d_{uf}} \quad (9)$$

To optimally complete a given desired task, the inequality constraints denoted by Eq. (9) should be integrated into the optimization problem. Of course, each link of the manipulator and every obstacle in the workspace, also between two different links yield inequality (9), which indicates that series of constraints will be activated according to the relative relationship of two arbitrary movable objects in the workspace.

4. Controller design

4.1. General NMPC strategy

Stemmed from the chemical processing industries, MPC has gradually expanded its applications into other

disciplines. The fundamental idea of the MPC is using a prediction model and numerical on-line optimization methods to obtain a sequence of control input that minimizes a predefined cost function over a finite time horizon, while subjects to certain constraints concurrently. At each sample time, concerning the new measurements and the current system states, prediction and optimization are performed to determine the optimal control input over the control horizon. The first control input will be adopted and such process will be repeated at the next sampling instant in a “receding horizon” manner (Fig. 3). From a theoretical point of view, the general NMPC strategy can be described as follows

$$\mathbf{u} = \arg_{\mathbf{u}} \min \Gamma(k)$$

subject to

$$\begin{cases} \mathbf{x}(k|k) = \mathbf{x}_k \\ \mathbf{u}(k+j|k) = \mathbf{u}(k+N_c|k), j \geq N_c \\ \mathbf{x}(k+j+1|k) = \mathbf{f}_d(\mathbf{x}(k+j|k), \mathbf{u}(k+j|k)) \\ \mathbf{y}(k+j|k) = \mathbf{h}_d(\mathbf{x}(k+j|k), \mathbf{u}(k+j|k)) \\ \mathbf{y}_{min} \leq \mathbf{y}(k+j|k) \leq \mathbf{y}_{max} \\ \mathbf{u}_{min} \leq \mathbf{u}(k+j|k) \leq \mathbf{u}_{max} \\ \mathbf{g}(\mathbf{x}(k+j|k)) \leq \mathbf{0} \end{cases} \quad (10)$$

where $j \in [0, N_p - 1]$, N_p and N_c are the prediction and control horizon, respectively. $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{u}$ represent the system states, output and control input, respectively. The notation $\mathbf{a}(i|j)$ describes the value of vector $\mathbf{a}$ at the instant i predicted at the instant j ($j < i$). The two functions, $\mathbf{f}_d(\mathbf{x}(k+j|k), \mathbf{u}(k+j|k))$ and $\mathbf{h}_d(\mathbf{x}(k+j|k), \mathbf{u}(k+j|k))$ stand for the discrete prediction model and measurement model, while the control input $\mathbf{u}$ and system output $\mathbf{y}$ yield to their corresponding lower and upper bounds. The inequality $\mathbf{g}(\mathbf{x}(k+j|k)) \leq \mathbf{0}$ represents the additional constraints, such as the security constraints, terminal constraints, and so forth. The cost function $\Gamma(k)$ is a scalar function which is usually determined in terms of the control effort and the difference between the predictive and the desired output. The optimization procedure must be solved at each sampling time k to obtain a sequence of optimal control input as $\{\mathbf{u}^*(k|k), \dots, \mathbf{u}^*(k+N_c-1|k)\}$. When different constraints are taken into account during the optimization, a constrained quadratic programming procedure can be employed to gain the optimal control input.

4.2. Optimization index

According to the NMPC strategy described in Eq. (10), an appropriate cost function $\Gamma(k)$ must be chosen to obtain the local optimal control law. Generally, the cost function $\Gamma(k)$ not only should include direct relation with the tracking error between the predictive controlled output $\hat{\mathbf{y}}(k+i|k)$ and the reference trajectory $\mathbf{r}(k+i|k)$, but also should include the control input effort $\Delta \hat{\mathbf{u}}(k+i|k)$ in order to save power/fuel consumption in optimization of

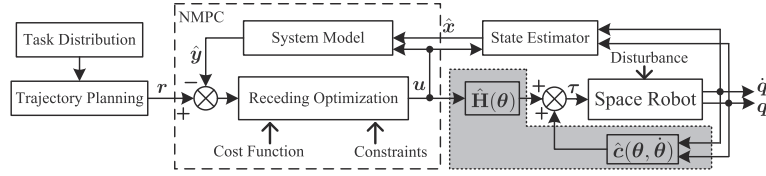


Fig. 3. Block scheme of the NMPC.

manoeuvres of the arm. Then a cost function with quadratic form can be described by

$$\Gamma(k) = \sum_{i=1}^{N_p} \|\hat{\mathbf{y}}(k+i|k) - \mathbf{r}(k+i|k)\|_{\mathbf{Q}(i)}^2 + \sum_{i=0}^{N_c-1} \|\Delta \hat{\mathbf{u}}(k+i|k)\|_{\mathbf{T}(i)}^2 \quad (11)$$

where $\mathbf{Q}(i)$ and $\mathbf{T}(i)$ are the weight matrices of the tracking error and control effort, respectively. If we define $\mathbf{Y}(k) = [\hat{\mathbf{y}}(k+1|k), \dots, \hat{\mathbf{y}}(k+N_p|k)]^T$, $\mathbf{R}(k) = [\mathbf{r}(k+1|k), \dots, \mathbf{r}(k+N_p|k)]^T$, and $\Delta \mathbf{U}(k) = [\Delta \hat{\mathbf{u}}(k|k), \dots, \Delta \hat{\mathbf{u}}(k+N_c-1|k)]^T$, then the cost function $\Gamma(k)$ can be written as follows

$$\Gamma(k) = \|\mathbf{Y}(k) - \mathbf{R}(k)\|_{\mathbf{Q}}^2 + \|\Delta \mathbf{U}(k)\|_{\mathbf{T}}^2 \quad (12)$$

where $\mathbf{Q} = \text{diag}([\mathbf{Q}(1), \dots, \mathbf{Q}(N_p)])$ and $\mathbf{T} = \text{diag}([\mathbf{T}(1), \dots, \mathbf{T}(N_c-1)])$. Here if we define

$$\mathbf{Y}(k) = \Phi \mathbf{x}(k) + \Upsilon \mathbf{u}(k-1) + \Theta \Delta \mathbf{U}(k) \quad (13)$$

Consider the system discrete model expressed in Eq. (5), the matrices Φ , Υ and Θ can be expressed by

$$\Phi = \begin{bmatrix} \mathbf{C}_d \mathbf{A}_d \\ \vdots \\ \mathbf{C}_d \mathbf{A}_d^{N_c} \\ \mathbf{C}_d \mathbf{A}_d^{N_c+1} \\ \vdots \\ \mathbf{C}_d \mathbf{A}_d^{N_p} \end{bmatrix}, \quad \Upsilon = \begin{bmatrix} \mathbf{C}_d \mathbf{B}_d \\ \vdots \\ \sum_{i=0}^{N_c-1} \mathbf{C}_d \mathbf{A}_d^i \mathbf{B}_d \\ \sum_{i=0}^{N_c} \mathbf{C}_d \mathbf{A}_d^i \mathbf{B}_d \\ \vdots \\ \sum_{i=0}^{N_p-1} \mathbf{C}_d \mathbf{A}_d^i \mathbf{B}_d \end{bmatrix} \quad (14)$$

$$\Theta = \begin{bmatrix} \mathbf{C}_d \mathbf{B}_d & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \sum_{i=0}^{N_c-1} \mathbf{C}_d \mathbf{A}_d^i \mathbf{B}_d & \cdots & \mathbf{C}_d \mathbf{B}_d \\ \sum_{i=0}^{N_c} \mathbf{C}_d \mathbf{A}_d^i \mathbf{B}_d & \cdots & \sum_{i=0}^1 \mathbf{C}_d \mathbf{A}_d^i \mathbf{B}_d \\ \vdots & \ddots & \vdots \\ \sum_{i=0}^{N_p-1} \mathbf{C}_d \mathbf{A}_d^i \mathbf{B}_d & \cdots & \sum_{i=0}^{N_p-N_c} \mathbf{C}_d \mathbf{A}_d^i \mathbf{B}_d \end{bmatrix} \quad (15)$$

In consideration of Eq. (13), let us define a new variable $\varepsilon(k)$ as follows

$$\varepsilon(k) = \mathbf{R}(k) - \Phi \mathbf{x}(k) - \Upsilon \mathbf{u}(k-1) \quad (16)$$

then the cost function $\Gamma(k)$ can be denoted by the following equation

$$\begin{aligned} \Gamma(k) &= \|\Theta \Delta \mathbf{U}(k) - \varepsilon(k)\|_{\mathbf{Q}}^2 + \|\Delta \mathbf{U}(k)\|_{\mathbf{T}}^2 \\ &= \Gamma_{\text{const}} + \Delta \mathbf{U}(k)^T \vartheta + \Delta \mathbf{U}(k)^T \mathbf{M} \Delta \mathbf{U}(k) \end{aligned} \quad (17)$$

where $\Gamma_{\text{const}} = \varepsilon(k)^T \mathbf{Q} \varepsilon(k)$, $\vartheta = -2\Theta^T \mathbf{Q} \varepsilon(k)$ and $\mathbf{M} = \Theta^T \mathbf{Q} \Theta + \mathbf{T}$.

4.3. Inequality constraints

Actually, all processes are subject to certain constraints. Sensors have their own limited scopes, and actuators have limited field of action. Furthermore, security, environmental and operational conditions are often defined by the intersection of certain constraints for safety or economic reasons. Therefore, the control strategy will be implemented under a list of constraints. These facts make the introduction of constraints in the cost function to be minimized necessary. In view of the cost function expressed in Eq. (17), since Γ_{const} is a scalar quantity, hence, the optimal control issue with the aforementioned constraints over the receding horizon can be denoted as follows

$$\begin{aligned} \Delta \mathbf{U}^*(k) &= \min_{\Delta \mathbf{U}} \Delta \mathbf{U}(k)^T \mathbf{M} \Delta \mathbf{U}(k) + \vartheta^T \Delta \mathbf{U}(k) \\ \text{subject to} \quad &\mathbf{G} \Delta \mathbf{U}(k) \leq \mathbf{g} \end{aligned} \quad (18)$$

which is a constrained quadratic optimization problem. Next, we will concentrate on translating various constraints into inequality equations and obtaining the matrix $\mathbf{G}$ and vector $\mathbf{g}$.

Generally, bounds in the amplitude of the control input $\tau(t)$ and limits in the output $\mathbf{y}(t)$ will be taken into account

$$\begin{cases} \tau_{\min} \leq \tau(t) \leq \tau_{\max} & \forall t \\ \mathbf{y}_{\min} \leq \mathbf{y}(t) \leq \mathbf{y}_{\max} & \forall t \end{cases} \quad (19)$$

Considering the system dynamics in Eq. (4) and the control input limits in Eq. (19), there is no linear relationship between the new input vector $\mathbf{u}(t)$ and real control input $\tau(t)$. Thanks to the positive symmetric inertia matrix $\mathbf{H}(\theta)$, we can define $\mathbf{u}_{\min} = \hat{\mathbf{H}}^{-1}(\underline{\tau} - \hat{\mathbf{c}})$ and $\mathbf{u}_{\max} = \hat{\mathbf{H}}^{-1}(\bar{\tau} - \hat{\mathbf{c}})$, where $\underline{\tau}$ and $\bar{\tau}$ are the corresponding torque vectors fulfil the control input boundary. The new input vector $\mathbf{u}(k)$ yields its own amplitude boundaries can be expressed by

$$\begin{cases} \Omega_{N_c} \mathbf{u}(k-1) + \Psi \Delta \mathbf{U}(k) \leq \Omega_{N_c} \mathbf{u}_{max} \\ \Omega_{N_c} \mathbf{u}(k-1) + \Psi \Delta \mathbf{U}(k) \geq \Omega_{N_c} \mathbf{u}_{min} \end{cases} \quad (20)$$

where the matrices Ω_{N_c} and Ψ can be described as follows

$$\Omega_{N_c} = \begin{bmatrix} \mathbf{E}_n \\ \vdots \\ \mathbf{E}_n \end{bmatrix} \in \mathbb{R}^{nN_c \times n}, \quad \Psi = \begin{bmatrix} \mathbf{E}_n & & \\ \vdots & \ddots & \\ \mathbf{E}_n & \cdots & \mathbf{E}_n \end{bmatrix} \in \mathbb{R}^{nN_c \times nN_c} \quad (21)$$

where $\mathbf{E}_n \in \mathbb{R}^{n \times n}$ is an n dimensional identity matrix. Likewise, suppose $\mathbf{Y}_p = \Phi \mathbf{x}(k) + \Upsilon \mathbf{u}(k-1)$ and $\Omega_{N_p} = [\mathbf{E}_{2n}, \dots, \mathbf{E}_{2n}]^T \in \mathbb{R}^{2nN_p \times 2n}$, the new input vector $\mathbf{u}(k)$ yields the boundaries of output $\mathbf{y}(t)$ can be denoted as

$$\begin{cases} \mathbf{Y}_p + \Theta \Delta \mathbf{U}(k) \leq \Omega_{N_p} \mathbf{y}_{max} \\ \mathbf{Y}_p + \Theta \Delta \mathbf{U}(k) \geq \Omega_{N_p} \mathbf{y}_{min} \end{cases} \quad (22)$$

As illustrated in Section 3, when the configuration of the space robot at one point slides into the influence area of an obstacle, a corresponding anti-collision constraint will be activated as expressed in Eq. (8). Assuming that N_k anti-collision constraints are activated at the instant k , the new input vector $\mathbf{u}(k)$ yields the velocity-damper constraint in Eq. (9) can be expressed by

$$\mathbf{D}_{vp} + \Theta_v \Delta \mathbf{U}(k) \geq \mathbf{D}_{uf} \quad (23)$$

where $\mathbf{D}_{vp} = \Theta_{vj} \mathbf{y}_{vp}$ and $\Theta_v = \Theta_{vj} \Theta_{vb}$. Here we choose $\mathbf{C}_{vd} = [\mathbf{0}_n, \mathbf{E}_n]$ and define $\mathbf{y}_{vp} = \mathbf{C}_{vd} \mathbf{A}_d \mathbf{x}(k) + \mathbf{C}_{vd} \mathbf{B}_d \mathbf{u}(k)$ and $\Theta_{vb} = [\mathbf{C}_{vd} \mathbf{B}_d, \mathbf{0}_n, \dots, \mathbf{0}_n] \in \mathbb{R}^{n \times nN_c}$, in terms of the inequality (9), Θ_{vj} and $\mathbf{D}_{uf}$ can be expressed as follows

$$\Theta_{vj} = \begin{bmatrix} \mathbf{n}_1^T \mathbf{J}_{P_1} \\ \vdots \\ \mathbf{n}_i^T \mathbf{J}_{P_i} \\ \vdots \\ \mathbf{n}_{N_k}^T \mathbf{J}_{P_{N_k}} \end{bmatrix}, \quad \mathbf{D}_{uf} = \begin{bmatrix} \langle \dot{\mathbf{r}}_{P_1}, \mathbf{n}_1 \rangle - \zeta \frac{d_{11'} - d_{uf}}{d_{1f} - d_{uf}} \\ \vdots \\ \langle \dot{\mathbf{r}}_{P_i}, \mathbf{n}_i \rangle - \zeta \frac{d_{ii'} - d_{uf}}{d_{if} - d_{uf}} \\ \vdots \\ \langle \dot{\mathbf{r}}_{P_{N_k}}, \mathbf{n}_{N_k} \rangle - \zeta \frac{d_{N_k N_k'} - d_{uf}}{d_{N_k f} - d_{uf}} \end{bmatrix} \quad (24)$$

When a variety of constraints are expressed by the inequality equations as in Eqs. (20), (22) and (23), the quantities of $\mathbf{G}$ and $\mathbf{g}$ can be derived as follows

$$\mathbf{G} = \begin{bmatrix} \Psi \\ -\Psi \\ \Theta \\ -\Theta \\ -\Theta_v \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} \Omega_{N_c} \mathbf{u}_{max} - \Omega_{N_c} \mathbf{u}(k-1) \\ -\Omega_{N_c} \mathbf{u}_{min} + \Omega_{N_c} \mathbf{u}(k-1) \\ \Omega_{N_p} \mathbf{y}_{max} - \mathbf{Y}_p \\ -\Omega_{N_p} \mathbf{y}_{min} + \mathbf{Y}_p \\ \mathbf{D}_{vp} - \mathbf{D}_{uf} \end{bmatrix} \quad (25)$$

After integrating various constraints into the cost function $\Gamma(k)$ and expressing the optimization issue in the standard form with linear inequalities as in Eq. (18), standard constrained QP searching method as introduced in Vada et al. (2001) or Wang (2009, Ch. 2) can be applied to search

the optimal solution $\Delta \mathbf{U}^*(k)$. The first optimal control input effort $\Delta \hat{\mathbf{u}}^*(k|k)$ in the vector $\Delta \mathbf{U}^*(k)$, together with the former control input $\mathbf{u}^*(k-1)$, will determine the current optimal control input $\mathbf{u}^*(k) = \mathbf{u}^*(k-1) + \Delta \hat{\mathbf{u}}^*(k|k)$. $\mathbf{u}^*(k)$ will be adopted as the optimal control input to exert on the system model and the constrained QP procedure will be repeated at the next sampling instant $k+1$ to obtain a new optimal control input $\mathbf{u}^*(k+1)$ in a receding horizon manner.

5. Simulation experiments

5.1. Implementation

This section illustrates the results of the simulations designed to demonstrate the viability and effectiveness of the proposed NMPC strategy. The designed NMPC acts as a multiple-input, multiple output (MIMO) system. The space robotic system used in this section is composed of a 6 DOF spacecraft and a 7 DOF kinematically redundant manipulator as shown in Fig. 4. The mass and inertia properties of the space robot are listed in Table 2, where the position vectors $\mathbf{a}_i$ and $\mathbf{b}_i$, mass m_i and inertia matrix $\mathbf{I}_i$ (defined in Table 1) of link i are expressed in its own body frame. In order to verify the effectiveness of the proposed NMPC, two simulation cases will be investigated by numerical simulations. The simulations are performed within Matlab/Simulink[®] environment.

The tuning parameters relevant to the NMPC are the sampling period h , prediction horizon N_p , control horizon N_c and the input and output weights matrix $\mathbf{Q}(i)$ and $\mathbf{T}(i)$. The value of h , N_p and N_c should be chosen according to the practical computational resources. The influence of h , N_p and N_c to the control performance can be found in Boscaroli et al. (2010). The computational cost of each step is directly proportional to both N_p and N_c . In this study, all the simulations were performed within 10 ms which indicates that $h = 0.01$ s. Due to the inaccuracy of long-term

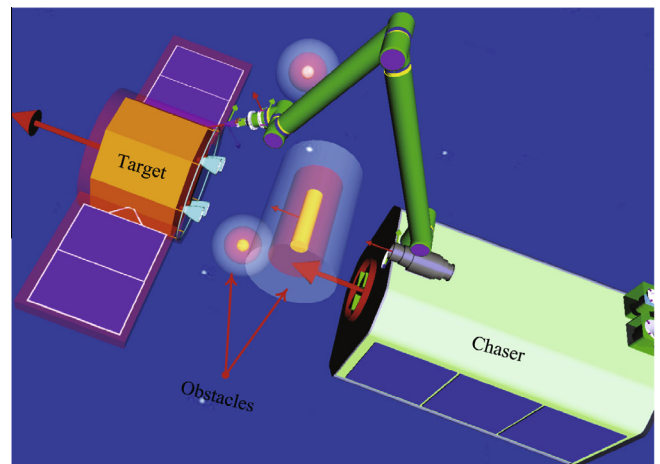


Fig. 4. Simulation snapshot of the space robot.

Table 2

Mass and inertia parameters of the space robot.

	Base	Link1	Link2	Link3	Link4	Link5	Link6	Link7
m (kg)	200	4.0	8.0	2.0	6.0	2.0	2.0	5.0
I_{xx} (kg m ²)	50	0.3	0.8	0.1	0.7	0.1	0.1	0.4
I_{yy} (kg m ²)	100	0.3	0.5	0.1	0.4	0.1	0.1	0.4
I_{zz} (kg m ²)	100	0.2	0.8	0.1	0.7	0.1	0.1	0.2
a_x (m)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
a_y (m)	0.0	0.0	−0.7	0.0	−0.5	0.0	0.0	0.0
a_z (m)	0.0	0.128	0.084	0.09	0.084	0.09	0.084	0.12
b_x (m)	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0
b_y (m)	0.084	0.084	−0.66	−0.084	−0.46	0.084	−0.09	0.0
b_z (m)	−0.6	0.128	0.0	0.0	0.0	0.0	0.0	0.21

prediction and high burden of digital calculation, the prediction horizon and control horizon were set to $N_p = 20$ and $N_c = 5$, respectively. The ratio between weight matrices $\mathbf{Q}(i)$ and $\mathbf{T}(i)$ indicates the relative significance of the tracking error and control effort. Here, we choose $\mathbf{Q}(i) = \mathbf{E}_{2n}$ and $\mathbf{T}(i) = 0.05\mathbf{E}_n$, which reflects that the tracking error is the primary optimization index with higher importance.

5.2. Case study for comparison with RMAC

In this simulation, a RMAC controller is implemented and tested: the simulation results are employed to compare the designed NMPC strategy with a commonly used

RMAC strategy. In the first simulation, we assume that there is no obstacles in the workspace of the manipulator. The required robotic mission is to unfold the manipulator from its initial configuration to an optimal grasping pose. The space manipulator is initially at its folding configuration $\theta_0 = [0, -\frac{\pi}{2}, 0, 0, 0, 0, 0]^T$. After receiving the unfolding command, the motors of the joints will be activated and will deploy the manipulator to a specific position $\theta_d = [0, \frac{\pi}{4}, 0, \frac{2\pi}{5}, 0, -\frac{\pi}{6}, 0]^T$ in 4 s ready for further robotic manipulation. For each joint motion, a quintic polynomial trajectory $r(t) = \sum_{i=0}^5 \beta_i (t - t_0)^i$ is employed as the set point, where β_i represents the coefficient vector of the planned trajectory based on inverse kinematics and the duration of the trajectory.

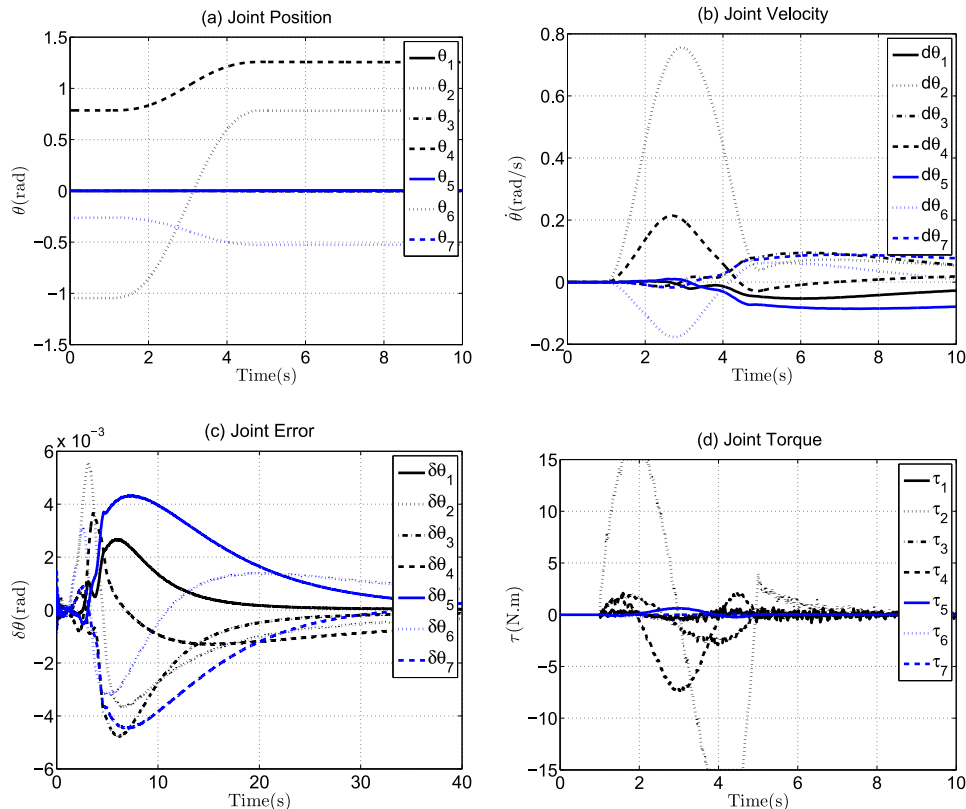


Fig. 5. Unfolding the space manipulator using RMAC.

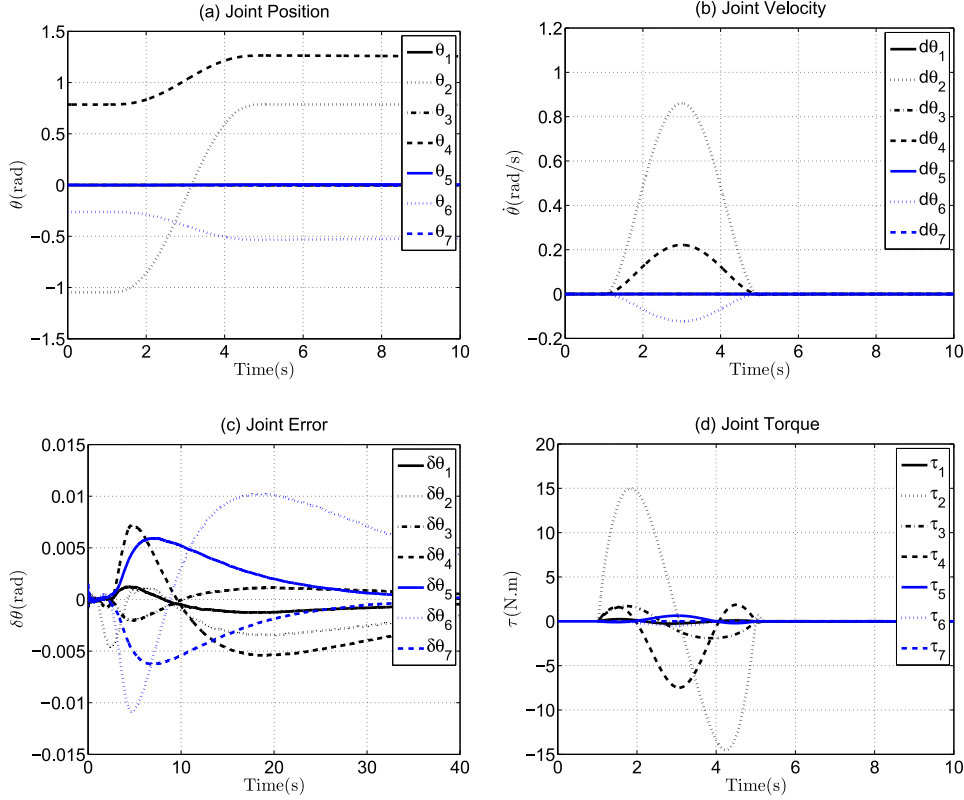


Fig. 6. Unfolding the space manipulator using NMPC.

Figs. 5 and 6 show the simulation results of the RMAC and NMPC methods, respectively. One can see that, both methods can complete the unfolding task successfully, while NMPC strategy holds extra advantages. The residue error of the joint position and velocity by using RMAC are larger than that of using NMPC. Besides, joint torque violates its range limits when using RMAC and have a noticeable overshoot ($\approx 20\%$). It can be clearly seen that the NMPC controller has a superior constraints handling capability.

5.3. Case study for path tracking with collision avoidance

In this simulation, in order to test the capacity of the proposed NMPC strategy with collision avoidance, as shown in Fig. 4, two spheres and one cylinder are artificially treated as obstacles and located in the workspace of the space manipulator. The parameters of d_{if} , d_{sr} , d_{uf} and ξ are respectively set to 0.2 m, 0.10 m, 0.05 m and 0.18. The primary task of the end-effector is to track a path with the form of a lemniscate in a plane within 10 s which is defined as follows

$$\begin{cases} \dot{x}_e = \omega_e^x = \omega_e^z = 0; \\ \dot{y}_e = -0.16\pi \sin(\frac{\pi t}{5} + \frac{\pi}{2}) \\ \dot{z}_e = 0.2\pi \cos(\frac{2\pi t}{5}) \\ \omega_e^y = 0.05 \sin(\frac{\pi t}{10}) \end{cases} \quad (26)$$

Unit quaternion $\mathbf{q} = \{\eta, \boldsymbol{\varepsilon}\} \in \mathbb{R}^4$ is used here to represent the orientation of the end-effector. η is the scalar part and $\boldsymbol{\varepsilon}$ is the vector part of the quaternion. Considering the forward kinematics in Eq. (3), the required joint velocity can be derived employing pseudo-inverse and null-space of the GJM $\mathbf{J}_g$ and $\dot{\boldsymbol{\theta}} = \mathbf{J}_g^+ \dot{\mathbf{x}}_e + (\mathbf{E}_n - \mathbf{J}_g^+ \mathbf{J}_g) \boldsymbol{\psi}$, where $\boldsymbol{\psi} \in \mathbb{R}^n$ is an arbitrary vector. Possible dynamic singularities may emerge during the end-effector's path tracking phase which can be overcome by the damped least squares strategy as developed in Chiaverini (1997).

The simulation results are shown in Fig. 7. One can see that, the path tracking task is not completely conducted since the existence of the obstacles. When the manipulator approaches to the obstacles, the anti-collision constraints are activated to impede the occurrence of possible collision. The minimum distances between each link of the manipulator and the obstacles are strictly constrained by the predefined unsafe distance d_{uf} . During the path tracking task, three anti-collision constraints were activated. After the manipulator escaping from the influence zone of the obstacles, the NMPC strategy continued to conduct the required task as much as possible. The joint position and velocity also fulfil the output constraints. The simulation results demonstrate that the proposed NMPC strategy is satisfied for obstacle avoidance especially when more than one obstacles in the workspace of the manipulator.

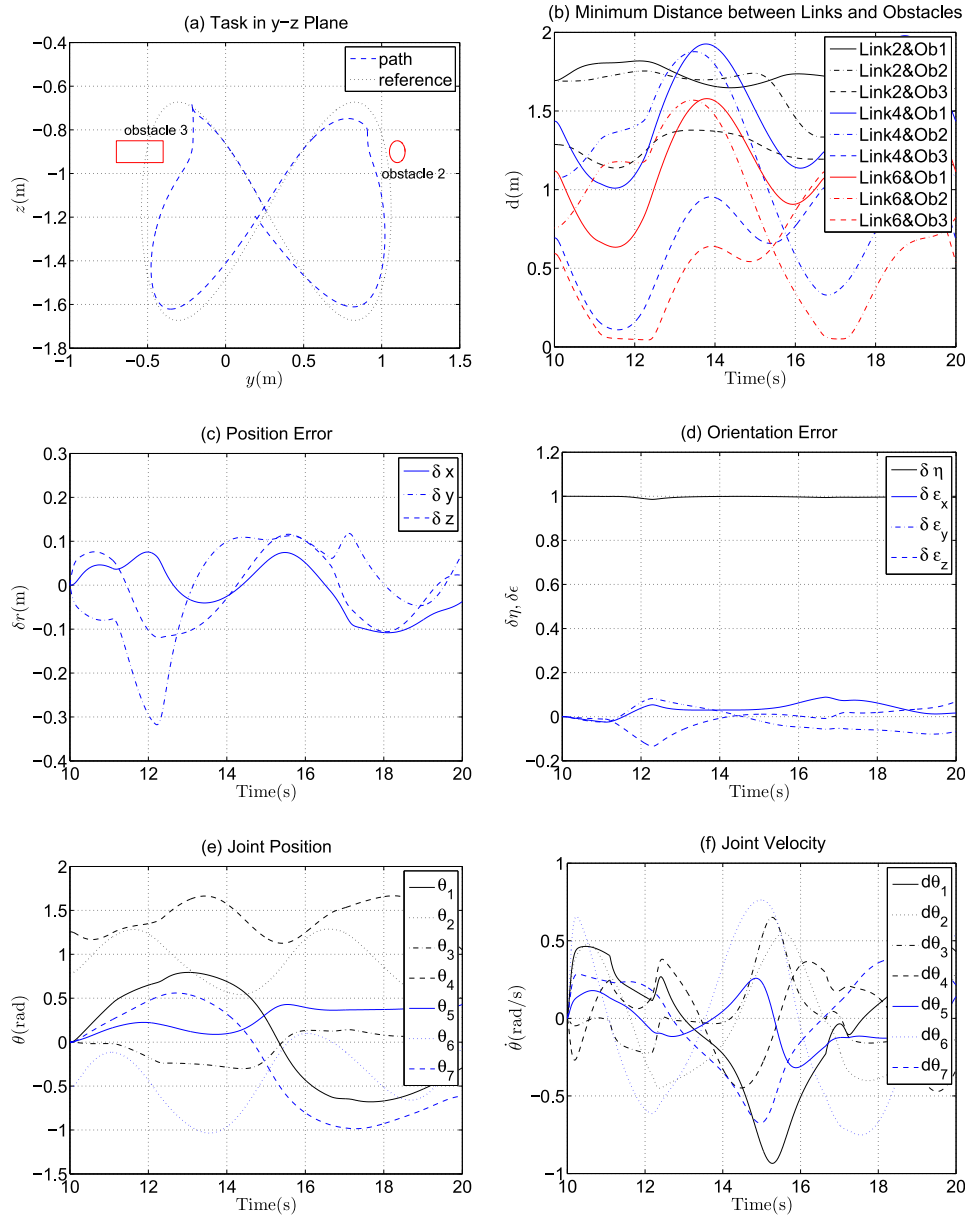


Fig. 7. Tracking a lemniscate with anti-collision constraints.

6. Conclusions

In this study, in order to control the motion of a space manipulator in its workspace with obstacles, a non-linear model predictive control (NMPC) strategy with obstacle avoidance has been designed and applied for the free-floating space robot. Collision avoidance issue was converted into linear inequality constraints which were integrated into quadratic programming (QP) optimization problem as part of NMPC strategy. The proposed algorithm coincides with human perception of dealing with such problems. An obstacle would not be taken into account until it slides into certain range. Moreover, the new algorithm can indisputably deal with multi-obstacles

and multi-links situation in the workspace by simply enforcing inequality constraints into the NMPC strategy. Two simulations, point-to-point approaching and predefined path tracking with multiple obstacles in the workspace have evidently demonstrated the feasibility and effectiveness of the proposed controller strategy. However, there are still space to improve in our study. One is the problem of strictly convex obstacles which is not always the case in practice. Another is the predetermined weight matrix in cost function. A better way would be adopting adjustable weight matrix as the space scenario changes. Future research in this field will include non-convex obstacles detection and adaptive NMPC strategy design.

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